A physical cut: elastic reversal, finite speed and memory

A05, reviewed with B13, 2026-09-07. An equal-mass elastic collision realizes a random midpoint with exact energy and momentum conservation. Its action cost tends to zero with the interval duration at fixed speed. A sharp midpoint bound extends this conclusion to every bounded-speed bridge law. Finally, a position-only convolution law with ballistic support is necessarily a deterministic drift. Together these tests identify velocity memory as a concrete ingredient for the next refinement model.

1. One collision with a momentum receiver

Fix $m > 0$, $0 < u < c$ and duration $\Delta > 0$. Choose a common random sign $\sigma = \pm 1$ with equal probabilities. At time zero, a labelled tracer is at 0 with velocity σu, and a second particle of the same mass is at $\sigma u \Delta$ with velocity $-\sigma u$. Both move freely on the line until their instantaneous elastic collision at time $\Delta/2$. The equal-mass collision exchanges their velocities. Thus the tracer follows

$$X(t) = \begin{cases} \sigma u t, & 0 \le t \le \Delta/2, \\ \sigma u(\Delta - t), & \Delta/2 \le t \le \Delta. \end{cases}$$

Its midpoint $Y = \sigma u \Delta/2$ is random, while $X(0) = X(\Delta) = 0$. The other particle returns to its own initial position. Throughout the motion, total momentum is zero and total kinetic energy is mu^2. Each labelled velocity changes sign at the collision, with opposite impulses of magnitude $2mu$. Both particles stay below c in the chosen frame.

The tracer's free kinetic action relative to the stationary path with the same position endpoints is

$$D = \frac{m}{2} \int_0^\Delta \dot{X}^2 dt = \frac{mu^2\Delta}{2}. \qquad \operatorname{Var} Y = \frac{u^2\Delta^2}{4}.$$

Define the midpoint action parameter in A03's normalization by $\kappa_{\mathrm{mid}} = 4m \operatorname{Var} Y/\Delta$. This gives $\kappa_{\mathrm{mid}} = mu^2\Delta$ and $\mathbb{E}D = \kappa_{\mathrm{mid}}/2$. At fixed energy and speed, both tend to zero as $\Delta \downarrow 0$. Keeping a supplied $\kappa_{\mathrm{mid}} > 0$ instead requires $u^2 = \kappa_{\mathrm{mid}}/(m\Delta)$ and pair energy $\kappa_{\mathrm{mid}}/\Delta$.

This is an exactly specified hard-collision model with random initial data. The timing follows from the prepared initial separation $u\Delta$, rather than a new stochastic clock. Changing Δ changes that separation. The stationary comparison shares positions and times, not initial velocities or apparatus state. Consequently the construction preserves the coarse position endpoints, but does not implement an invisible intervention on a fixed full phase-space preparation. Smooth collision potentials and a Lorentz-covariant dynamics are separate models.

2. Sharp midpoint support and variance bound

Let any absolutely continuous path on $[0, \Delta]$ have endpoints x, z and $|\dot{X}| \le u < c$ almost everywhere. Write $v = (z - x)/\Delta$, with $|v| \le u$, and $\zeta = X(\Delta/2) - (x + z)/2$. Each half-interval gives

$$X(\Delta/2) \in [x - u\Delta/2, x + u\Delta/2] \cap [z - u\Delta/2, z + u\Delta/2].$$

This intersection is centered at $(x + z)/2$ with half-width $r = \Delta(u - |v|)/2$. Therefore any probability law on these admissible paths satisfies

$$\operatorname{Var} Y \le r^2, \qquad 0 \le \kappa_{\mathrm{mid}} := \frac{4m}{\Delta} \operatorname{Var} Y \le m\Delta(u - |v|)^2.$$

Proof of the variance bound: $\operatorname{Var} Y \leq \mathbb{E}(Y-(x+z)/2)^2 \leq r^2$. Equal probabilities at the two extremal midpoints saturate it; the corresponding two-segment paths obey the speed bound. For $x = z$ the elastic example realizes this sharp case.

The two-segment interpolant through Y has excess kinetic action $D_{\text{poly}} = 2m\zeta^2/\Delta$. When the midpoint law is centered on $(x+z)/2$, $\mathbb{E}D_{\text{poly}} = \kappa_{\text{mid}}/2$. For a biased midpoint there is the additional term $2m(\mathbb{E}\zeta)^2/\Delta$. This distinction avoids identifying a variance with all action costs.

A Gaussian midpoint with variance $\kappa\Delta/(4m)$ has unbounded support for every $\kappa > 0$ and so fails the exact speed constraint at every duration. Even a bounded replacement matching only that variance requires $\kappa \leq m\Delta(u-|v|)^2$. The latter inequality gives a necessary resolution restriction, not a derivation of κ.

At $|v| = u$ the allowable midpoint is unique. Indeed a path achieving the maximum displacement $u\Delta$ must have velocity u almost everywhere. Thus after choosing an extremal midpoint in the return-path example, each half is already a saturated straight segment: finer admissible sampling inside that half is deterministic. Reapplying a nonzero midpoint fluctuation rule independently at every cut would change the coarse law or violate speed.

3. Finite speed and a position-only convolution law

Proposition. Suppose probability measures μ_t on the line satisfy $\mu_0 = \delta_0$, $\mu_{s+t} = \mu_s * \mu_t$, and $\operatorname{supp} \mu_t \subset [-ut, ut]$ for every $t \geq 0$, with a fixed finite u. Then $\mu_t = \delta_{bt}$ for some $|b| \leq u$.

Proof. For fixed $T > 0$ and every positive integer n, convolution and bounded support give

$$\operatorname{Var} \mu_T = n \operatorname{Var} \mu_{T/n} \leq nu^2(T/n)^2 = u^2T^2/n.$$

Let $n \to \infty$ to obtain zero variance at every T. Write $\mu_t = \delta_{a(t)}$. The convolution law makes a additive, while $|a(t)| \leq ut$ implies continuity at zero. Hence $a(t) = bt$, $|b| \leq u$. This proof even derives the needed continuity from the support bound.

The proposition concerns stationary, spatially homogeneous, independent increment composition on position alone. It leaves room for stochastic bounded-speed trajectories with velocity memory. For example A01's telegraph process is Markov in (X, V); deleting V removes the information needed for such position-only convolution. General state-dependent Markov motion is also outside the stated convolution hypothesis.

4. Consequence for the cut-point programme

Finite propagation does not supply a positive microscopic midpoint action coefficient in these models. It identifies a structural requirement: a stochastic refinement law with strict speed must retain correlations or additional state information, rather than resetting independent position increments. The elastic construction makes this visible at a single cut.

The next bounded task is to derive a finite-speed bridge on (X, V), including its conditional midpoint law and endpoint atoms, and compare coarse and fine restrictions of the same experiment. Then separate a finite observation-window action scale from a microscopic refinement remainder. The universal positive scale remains a selection question about the physical premises.

Proof and literature record

The B13 audit reviews C030–C032 as derived consequences. Barkai, *Stable Equilibrium Based on Lévy Statistics* (2003), arXiv:cond-mat/0303255v1, pp. 3–4, (2)–(3), supplies the elastic map. The equal-mass map is well-defined even though the divided-density representation (5) excludes that mass ratio. Cinque, *A Note on the Conditional Probabilities of the Telegraph Process* (2022), arXiv:2202.01904v1, pp. 1–2, supplies the velocity-memory and endpoint-atom comparison. The midpoint and convolution proofs above follow directly from their stated hypotheses;

neither selected source states those combined results. The source companion records the bounded coverage. No priority claim follows from this audit.